Surface Area

Exam code: 0607

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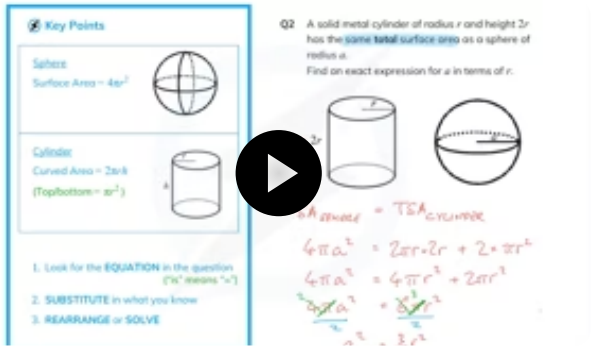

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Surface Area

What is surface area?

- The **surface area** of a 3D object is the **sum of the areas of all the faces** that make up the shape
  - Area is a 2D idea being applied into a 3D situation
  - A face is one of the flat or curved **surfaces** that make up a 3D object

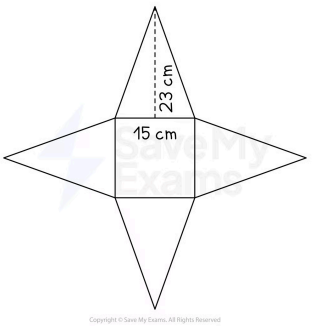
How do I find the surface area of cubes, cuboids, pyramids, and prisms?

- In cubes, cuboids, polygonal-based pyramids, and polygonal-based prisms (ie. pyramids and prisms whose bases have straight sides), **all the faces are flat**
- The **surface area** is found by
  - calculating the area of each individual flat face
  - adding these areas together
- You should remember the formula for the **area of a rectangle**, but you are given the **area of a triangle** in your exam
- When calculating surface area, it can be helpful to **draw a 2D net for the 3D shape** in question
  - For example, consider a square-based pyramid where the top of the pyramid is directly above the

Help me understand

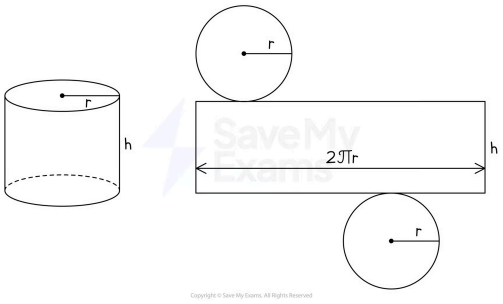
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Calculate the area of a square and the area of each triangle then add them together



How do I find the surface area of a cylinder?

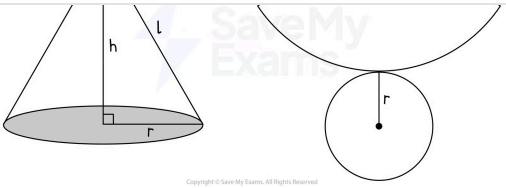
- A cylinder has **two flat surfaces** (the top and the base) and **one curved surface**
- The **net** of a cylinder consists of **two circles** and a **rectangle**



- The **curved surface area** of a cylinder, *A*, with base radius, *r*, and height, *h*, is therefore given by
  - $A = 2\pi r h$
  - This formula is given to you in the exam
- The **total surface area** of a cylinder, A_{Total}, can be found using the formula
  - $A_{Total} = 2\pi r h + 2\pi r^2$
  - This formula is **not** given to you in the exam

How do I find the surface area of a cone?

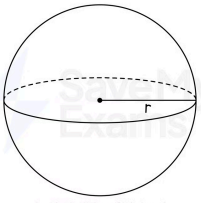
- A cone has **one flat surface** (the base) and **one curved surface**
- The net of a cone, with radius, *r*, perpendicular height, *h*, and sloping edge, (slant height), *l*, consists of
  - A **circular base**
  - A **sector** with radius, *l*, and an arc length equal to the circumference of the base



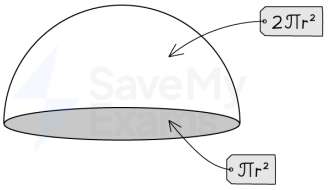
- The **curved surface area** of a cone, *A*, with radius, *r*, perpendicular height, *h*, and sloping edge, *l*, can be found using the formula
  - $A = \pi r l$
  - This formula is given to you in the exam
- The **total surface area** of a cone, A_{Total}, can be found using the formula
- $A_{Total} = \pi r l + \pi r^2$
- This formula is **not** given to you in the exam

How do I find the surface area of a sphere?

- A sphere has a **single curved surface**



- The surface area of a sphere, *A*, with radius, *r*, can be found using the formula
  - $A = 4\pi r^2$
  - This formula is given to you in the exam
- A **hemisphere** has half the curved surface area of a sphere and the flat circular base



- The surface area of a hemisphere, *A*, with radius, *r*, can be found using the formula
  - $A = 2\pi r^2 + \pi r^2$
  - This formula is **not** given to you in the exam

are given to you in the exam
- Read the question carefully, you may need to add additional areas, e.g. a base
- Make you are confident in calculating the areas of rectangles, circles and triangles

Worked Example

A toy consists of a cone of radius 5 cm and slant height 12 cm placed on top of a hemisphere with the same radius. Find the total surface area of the toy. Give your answer to 3 significant figures.

Answer:

Calculate the curved surface area of the cone using the formula, $A = \pi r l$

$$A_{cone} = \pi \times 5 \times 12 = 60\pi$$

Calculate the curved surface area of a hemisphere using the formula for the curved surface area of a sphere, $A = 4\pi r^2$ and dividing it by 2

$$A_{hemisphere} = \frac{4\pi \times 5^2}{2} = 50\pi$$

Add the two areas together to find the total surface area of the toy

$$A_{Totsl} = 60\pi + 50\pi = 110\pi$$

Evaluate on your calc and round to 3 significant figures

$$A_{Total} = 345.57519...$$

346 cm^2 (3 s.f.)

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